

Car Fleet Merging

There are n cars on a number line, numbered from 0 to $n - 1$. The i -th car is initially located at `position $i$` , and starts moving to the right at `speed $i$` units per second. The initial positions are strictly increasing, so `position $i$  < position $i+1$` for all i .

If a car catches up to another car (or a fleet of cars) ahead of it, they merge into a single fleet. A fleet moves at the speed of the slowest car within it (the *leader* of the fleet), and all cars in a particular fleet are located at the same position at any time.

You are given the `position` and `speed` arrays, each of length n . For each car, determine the number of the car it first catches up to. If a car first catches up to a fleet, answer with the number of that fleet's leader. If a car catches up to multiple cars or fleets simultaneously, answer with the number of the slowest car among them (as it becomes the leader of the newly formed fleet). If a car never catches up to any car ahead of it, answer with -1 .

Example Test Cases:

Test Case 1

Input: $n = 4$, `position` = [1,2,4,7], `speed` = [2,1,3,2]

Output: [1,-1,3,-1]

Explanation: Car 0 catches up to car 1. Car 2 catches up to car 3. Cars 1 and 3 never catch up to any cars ahead of them.

Test Case 2

Input: $n = 4$, `position` = [3,5,6,9], `speed` = [4,4,3,1]

Output: [3,2,3,-1]

Explanation: Car 1 first catches up to car 2, and they form a fleet. The fleet led by car 2 catches up to car 3, and car 3 becomes the new leader of the fleet. Car 0 eventually catches up to the fleet led by car 3.

Test Case 3

Input: $n = 5$, `position` = [2,6,8,11,14], `speed` = [5,4,2,4,1]

Output: [2,2,4,4,-1]

Explanation: Car 1 catches up to car 2, and car 3 catches up to car 4. Car 0 catches up to the fleet led by car 2. Eventually, the fleet led by car 2 catches up to the fleet led by car 4. Car 4 never catches up to any cars ahead of it.

Expected Time Complexity: $O(n)$

Input Format:

An integer n , an array `position` of length n , and an array `speed` of length n .

Output Format:

Return an array of n integers representing the caught car numbers, as defined above.

Constraints:

1. $1 \leq n \leq 10^5$
2. $1 \leq \text{position}_i, \text{speed}_i \leq 10^6$
3. $\text{position}_i < \text{position}_{i+1}$ for all i

Instructions:

You have been provided with two template files, containing the `main` and `solve` functions respectively. The `main` function, which has been completed for you, parses the input for each test case. For each test case, it calls the `solve` function, which must compute the answers.

The `solve` function must compute this in $O(n)$ time.

(Refer to the `Instructions.pdf` file for detailed guidelines on the input-output format and additional test cases.)

Library Usage

For this lab, you are permitted to use any modules which are part of the standard library of either C++ or python. For C++, do not add any extra `#include` directives; the provided `bits/stdc++.h` covers all allowed modules.